📊 Sales Data Analysis Report

1. Introduction

This project focuses on analyzing a retail dataset to understand sales performance, customer behavior, and product trends. The goal is to extract meaningful insights that can help improve business decisions.

---

2. Dataset Overview

The dataset contains transactional records of an online retail store with the following key attributes:

- Invoice Number
- Product Description
- Quantity
- Invoice Date
- Unit Price
- Customer ID
- Country

The data includes sales from multiple countries, with the majority of transactions from the United Kingdom.

---

3. Data Cleaning

The following steps were performed to prepare the data:

- Removed missing values from important columns
- Filtered out negative quantities (returns/cancellations)
- Converted date column into proper datetime format
- Created a new column **Revenue = Quantity × Unit Price**

---

4. Key Performance Indicators (KPIs)

- **Total Revenue:**

```
df['Revenue'].sum()
```

```
np.float64(8911407.904)
```

- **Top Product:**

```
top_products = df.groupby('Description')['Revenue'].sum().sort_values(ascending=False).head(10)
top_products
```

```
Description
PAPER CRAFT , LITTLE BIRDIE           168469.60
REGENCY CAKESTAND 3 TIER              142592.95
WHITE HANGING HEART T-LIGHT HOLDER    100448.15
JUMBO BAG RED RETROSPOT                85220.78
MEDIUM CERAMIC TOP STORAGE JAR         81416.73
POSTAGE                                77803.96
PARTY BUNTING                          68844.33
ASSORTED COLOUR BIRD ORNAMENT          56580.34
Manual                                 53779.93
RABBIT NIGHT LIGHT                     51346.20
Name: Revenue, dtype: float64
```

- **Top Country:** United Kingdom

```
top_countries = df.groupby('Country')['Revenue'].sum().sort_values(ascending=False).head(10)
top_countries
```

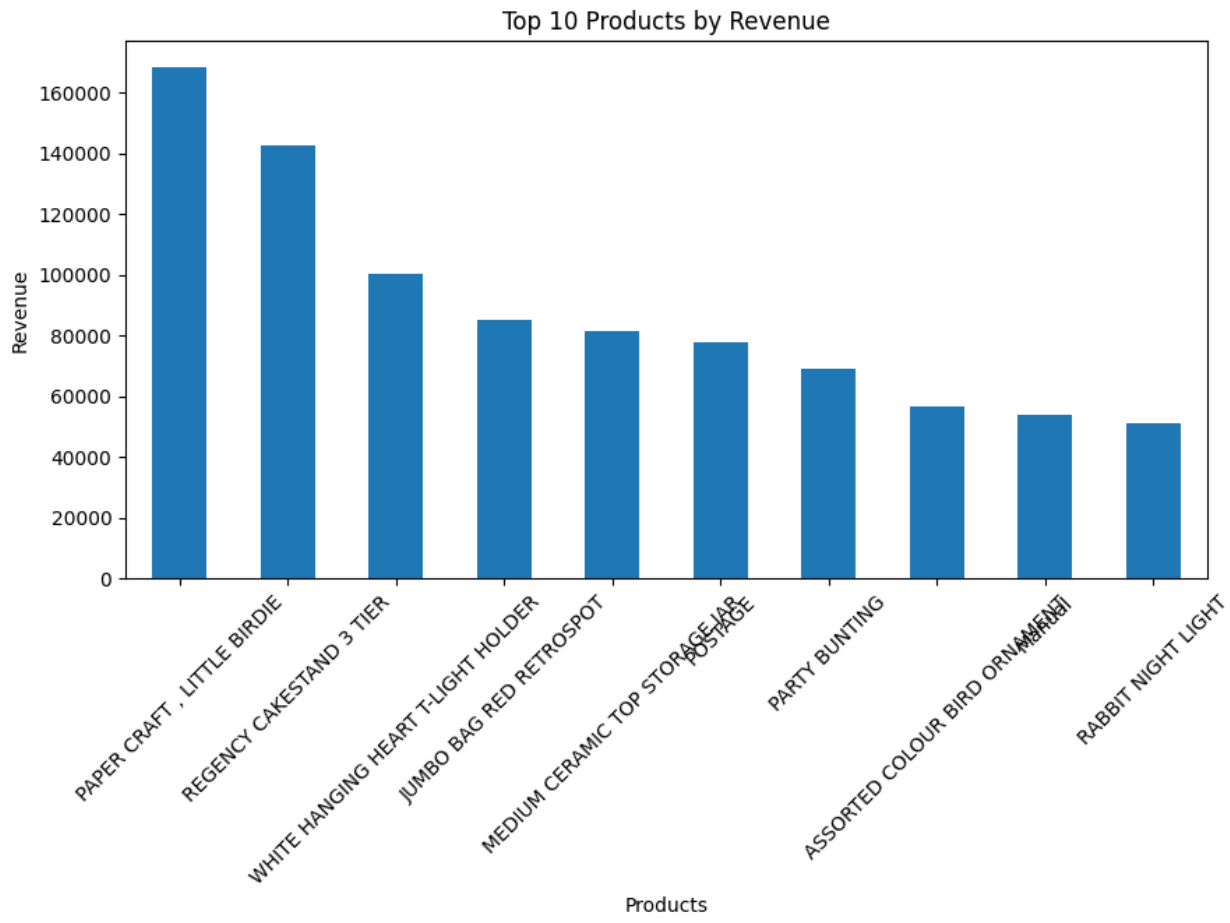
```
Country
United Kingdom    7308391.554
Netherlands        285446.340
EIRE               265545.900
Germany            228867.140
France             209024.050
Australia          138521.310
Spain               61577.110
Switzerland         56443.950
Belgium             41196.340
Sweden              38378.330
Name: Revenue, dtype: float64
```

---

5. Data Analysis & Visualization

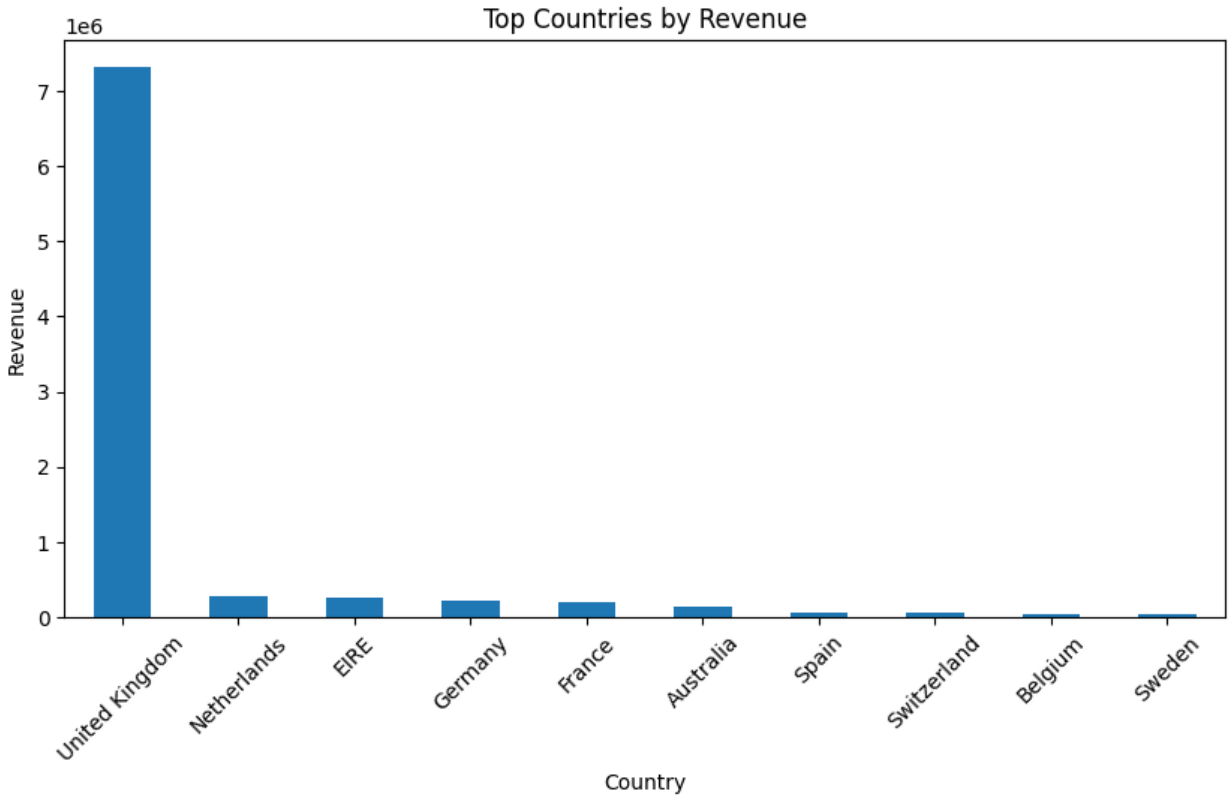
📌 Top Products

The analysis shows that a small number of products generate a large portion of total revenue.



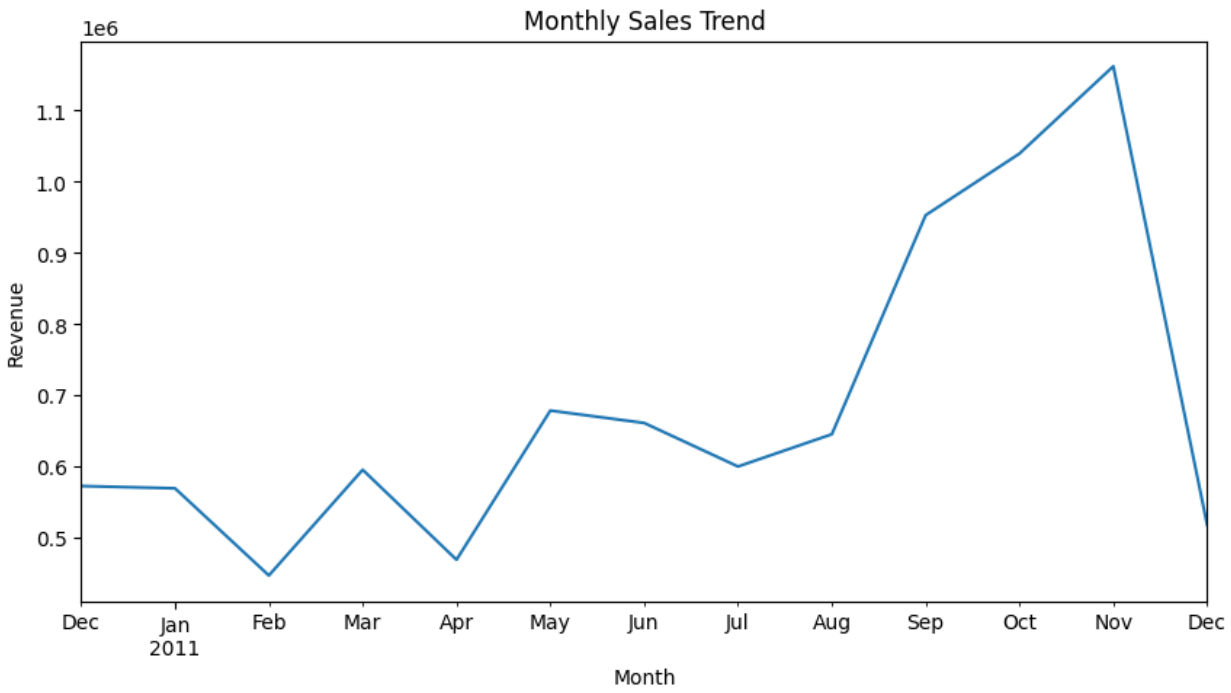
📌 Revenue by Country

The United Kingdom dominates sales compared to other countries.



📌 Monthly Sales Trend

Sales vary over time, indicating seasonal patterns and fluctuations.



---

6. Key Insights

- The **United Kingdom** is the highest revenue-generating country
- A limited number of products contribute significantly to overall sales
- Sales trends indicate possible seasonal demand patterns
- High-performing products should be prioritized for inventory management
- Low-performing regions can be targeted with marketing strategies

---

7. Conclusion

The analysis provides valuable insights into sales performance and customer trends. By focusing on top products and high-performing regions, the business can optimize revenue and improve decision-making.